DemandIQ

End-to-End Sales Forecasting & Demand Intelligence System

Anish Reddy | Data Science Intern Project

16.6%
Forecast MAPE
Weighted ensemble of 4 models

+29.9%
vs. Seasonal Baseline
Improvement over naive benchmark

8
Dashboard Pages
Live, interactive Streamlit app

793
Customers Segmented
RFM clustering, 4 segments

90%
Conformal Coverage
Model-agnostic prediction intervals

102
Notebook Cells
Zero execution errors, end-to-end

What's Inside

- 4 forecasting models benchmarked against 2 naive baselines (SARIMA, Prophet w/ holidays, XGBoost, weighted Ensemble)
- SHAP explainability + model-agnostic conformal prediction intervals (distribution-free uncertainty bounds)
- Dual-method anomaly detection AND K-Means clustering applied to both products and customers (RFM)
- Monte Carlo inventory cost simulator (newsvendor-style) + an A/B testing lab with power analysis
- Auto-generated executive briefing with one-click PDF export, live from the current data

Scan for the GitHub repo
https://anish1601171845-salesforecasting-anishreddy-app-8mfep4.streamlit.app/

Tech Stack

Python * Pandas/NumPy * XGBoost * Prophet * Statsmodels (SARIMA) * SHAP * scikit-learn * SciPy * Streamlit * Plotly